

Data Filtering DSL

The DSL allows creating and manipulating lists of JSON-like records with concise operators. You start by defining a list using **List(...)** where each element is a record with named fields. You can filter records using the **?** operator combined with conditions that support logical **and**. To transform records, you can add new fields with **+ field = value** and remove unwanted fields using **- field**. Selecting specific fields for output is done via **-> (field1, field2, ...)**, producing either a list of values or records containing only the chosen fields.

Control flow is supported with **if** statements, enabling conditional execution based on filter conditions, and **for** loops that iterate over lists to process or print each record individually. The **puts** statement outputs the result of expressions, whether a full list or individual records. Expressions support chaining these operations in sequence, allowing complex queries and transformations to be composed cleanly and readably.

The grammar enforces that expressions are list-centric, supporting chaining of filters, additions, removals, and selections. Conditions in filters allow simple comparisons and compound conditions with **and**. The DSL syntax does not require explicit line breaks or delimiters, relying on statement boundaries to separate commands. This design balances expressiveness and simplicity, making it straightforward to query, modify, and output JSON-style data within a compact, readable syntax.